Technical Document #993: Cybersecurity - Comprehensive Overview

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Vulnerability management identifies, classifies, and remediates security vulnerabilities. Regular scanning and patching are essential components.

Intrusion detection systems (IDS) monitor network traffic for suspicious activity. They alert administrators of potential security breaches.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Key Takeaways:

- Vulnerability management identifies, classifies, and remediates security vulnerabilities.
- Incident response plans define how organizations respond to security incidents.
- Encryption converts plaintext into ciphertext to protect data confidentiality.